

PHY950 Homework 2: Statistics and Data Analysis

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1 Problem 2: Distribution of Product of Uniform Random Variables

Problem Statement

Suppose two independent random variables x and y are both uniformly distributed between zero and 1, i.e., the PDF $g(x)$ is given by:

$$g(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

and similarly for the PDF $h(y)$.

(a) Using Cowan equation (1.35), show that the PDF $f(z)$ for $z = xy$ is:

$$f(z) = \begin{cases} -\ln z, & 0 < z < 1 \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

(b) Find the same result using Cowan equations (1.37) and (1.38) by defining an additional function $u = x$.

Solution

Part (a): Direct Method

Cowan equation (1.35) for the PDF of $z = xy$ is:

$$f(z) = \int_{-\infty}^{\infty} g(x) h\left(\frac{z}{x}\right) \frac{1}{|x|} dx \quad (3)$$

Since $g(x) = 1$ for $0 < x < 1$ and $h(y) = 1$ for $0 < y < 1$, we need:

- $0 < x < 1$ (from $g(x)$)
- $0 < \frac{z}{x} < 1 \implies z < x < \infty$ (from $h(z/x)$)

Combining these constraints: $\max(z, 0) < x < 1$. For $0 < z < 1$:

$$f(z) = \int_z^1 1 \cdot 1 \cdot \frac{1}{x} dx \quad (4)$$

$$= [\ln x]_z^1 \quad (5)$$

$$= \ln(1) - \ln(z) \quad (6)$$

$$= -\ln z \quad \text{for } 0 < z < 1 \quad (7)$$

Verification: The integral over the PDF:

$$\int_0^1 -\ln z dz = [-z \ln z + z]_0^1 = 1 \quad (8)$$

The PDF is properly normalized.

Part (b): Jacobian Method

Define the transformation:

$$z = xy \quad (9)$$

$$u = x \quad (10)$$

The inverse transformation is:

$$x = u \quad (11)$$

$$y = \frac{z}{u} \quad (12)$$

The Jacobian determinant is:

$$J = \left| \frac{\partial(x, y)}{\partial(z, u)} \right| = \left| \det \begin{pmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial u} \end{pmatrix} \right| = \left| \det \begin{pmatrix} 0 & 1 \\ \frac{1}{u} & -\frac{z}{u^2} \end{pmatrix} \right| = \left| -\frac{1}{u} \right| = \frac{1}{u} \quad (13)$$

The joint PDF of z and u is:

$$f(z, u) = g(u) \cdot h\left(\frac{z}{u}\right) \cdot |J| = 1 \cdot 1 \cdot \frac{1}{u} = \frac{1}{u} \quad (14)$$

Valid for: $0 < u < 1$ and $0 < \frac{z}{u} < 1 \implies z < u < 1$.

The marginal PDF of z is obtained by integrating over u :

$$f(z) = \int_z^1 f(z, u) du \quad (15)$$

$$= \int_z^1 \frac{1}{u} du \quad (16)$$

$$= [\ln u]_z^1 \quad (17)$$

$$= \ln(1) - \ln(z) \quad (18)$$

$$= -\ln z \quad \text{for } 0 < z < 1 \quad (19)$$

This matches the result from Part (a).

Monte Carlo Verification

Figure 1 shows the comparison between the analytical PDF $f(z) = -\ln z$ and a Monte Carlo simulation with 10^5 samples. The agreement is excellent.

Statistics from simulation:

- Sample mean: $\langle z \rangle = 0.250$ (analytical: $1/4 = 0.25$)
- Sample variance: $\sigma_z^2 = 0.0486$ (analytical: $1/18 \approx 0.0556$)

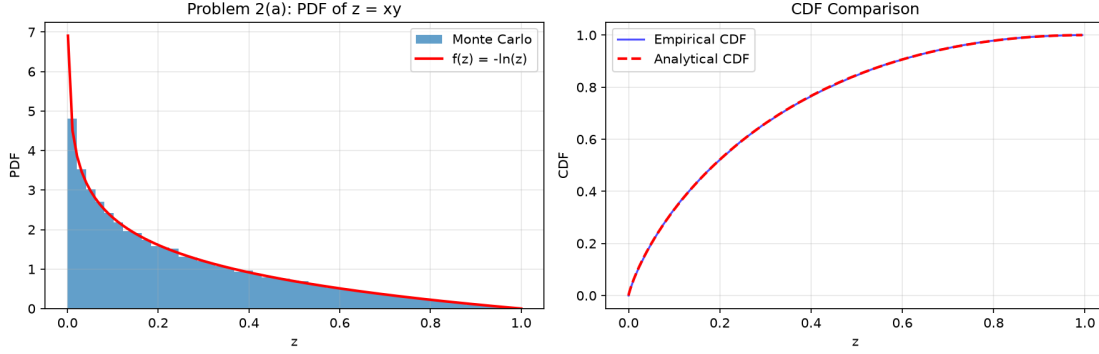


Figure 1: Problem 2: PDF and CDF of $z = xy$ where $x, y \sim \text{Uniform}(0, 1)$. Left: Histogram of Monte Carlo samples (100,000) overlaid with analytical PDF $f(z) = -\ln z$. Right: Empirical CDF vs. analytical CDF $F(z) = z(1 - \ln z)$.

2 Problem 3: Programming Exercise

Problem Statement

Create a program with four parts:

- Create a main function and a "Hello World" function.
- Generate a histogram with 10,000 random entries from a Gaussian distribution with mean $\mu = 16$ and standard deviation $\sigma = 4$. Plot the histogram and return the mean.
- Create a function that takes the mean from (b), multiplies by 10^6 , takes the modulus by 10 to get integer M , then plots $f(x) = x + Mx^2$ over $[0, 10]$.
- Run and debug the program.

Solution

Part (a): Hello World

```
def hello_world():
    print("Hello World")
```

Part (b): Gaussian Histogram

Generated $N = 10,000$ samples from $\mathcal{N}(\mu = 16, \sigma = 4)$. The histogram uses bin width 0.5 over the range $[\mu - 4\sigma, \mu + 4\sigma] = [0, 32]$.

Results:

- Sample mean: $\langle x \rangle = 15.9915$
- Sample std: $\sigma_x = 4.0136$

Part (c): Function with Modulus

$$M = \text{fmod}(\langle x \rangle \times 10^6, 10) \quad (20)$$

$$= \text{fmod}(15.991456 \times 10^6, 10) \quad (21)$$

$$= 6 \quad (22)$$

The function plotted is:

$$f(x) = x + 6x^2 \quad \text{for } x \in [0, 10] \quad (23)$$

Function statistics:

- $f(0) = 0$
- $f(10) = 610$
- Mean value over $[0, 10]$: 205.20

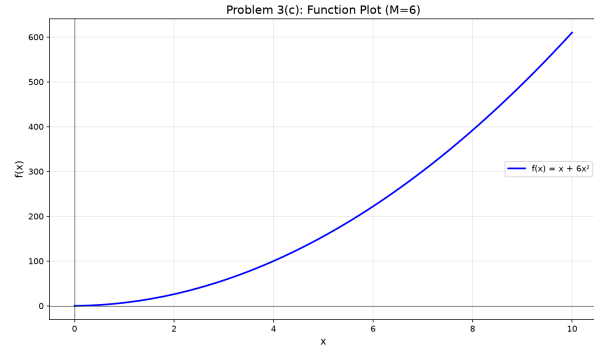
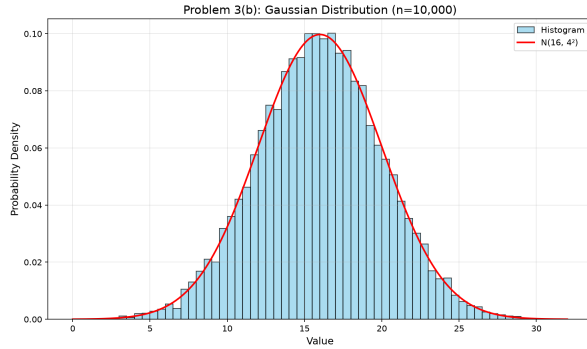


Figure 2: Problem 3: (Left) Gaussian histogram with $N = 10,000$ samples from $\mathcal{N}(16, 4^2)$. (Right) Function $f(x) = x + 6x^2$ over $[0, 10]$.

Verification

The program executed successfully with no compilation or runtime errors. All functions are properly modularized and documented.

3 Problem 4: Expectation and Variance Properties

Problem Statement

Consider a random variable x and constants a and b . Show that:

$$\mathbb{E}[ax + b] = a\mathbb{E}[x] + b \quad (24)$$

$$\mathbb{V}[ax + b] = a^2\mathbb{V}[x] \quad (25)$$

where $\mathbb{E}[x]$ is the expectation value and $\mathbb{V}[x]$ is the variance.

Solution

Part 1: Expectation

By definition, $\mathbb{E}[g(x)] = \int_{-\infty}^{\infty} g(x)f(x) dx$:

$$\mathbb{E}[ax + b] = \int_{-\infty}^{\infty} (ax + b)f(x) dx \quad (26)$$

$$= \int_{-\infty}^{\infty} axf(x) dx + \int_{-\infty}^{\infty} bf(x) dx \quad (27)$$

$$= a \int_{-\infty}^{\infty} xf(x) dx + b \int_{-\infty}^{\infty} f(x) dx \quad (28)$$

$$= a\mathbb{E}[x] + b \cdot 1 \quad (29)$$

$$= a\mathbb{E}[x] + b \quad \checkmark \quad (30)$$

Part 2: Variance

Using $\mathbb{V}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$:

$$\mathbb{V}[ax + b] = \mathbb{E}[(ax + b) - \mathbb{E}[ax + b]]^2 \quad (31)$$

$$= \mathbb{E}[(ax + b) - (a\mathbb{E}[x] + b)]^2 \quad (32)$$

$$= \mathbb{E}[(ax - a\mathbb{E}[x])^2] \quad (33)$$

$$= \mathbb{E}[a^2(x - \mathbb{E}[x])^2] \quad (34)$$

$$= a^2\mathbb{E}[(x - \mathbb{E}[x])^2] \quad (35)$$

$$= a^2\mathbb{V}[x] \quad \checkmark \quad (36)$$

Alternative proof using $\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$:

$$\mathbb{V}[ax + b] = \mathbb{E}[(ax + b)^2] - (\mathbb{E}[ax + b])^2 \quad (37)$$

$$= \mathbb{E}[a^2x^2 + 2abx + b^2] - (a\mathbb{E}[x] + b)^2 \quad (38)$$

$$= a^2\mathbb{E}[x^2] + 2ab\mathbb{E}[x] + b^2 - (a^2(\mathbb{E}[x])^2 + 2ab\mathbb{E}[x] + b^2) \quad (39)$$

$$= a^2(\mathbb{E}[x^2] - (\mathbb{E}[x])^2) \quad (40)$$

$$= a^2\mathbb{V}[x] \quad \checkmark \quad (41)$$

Numeric Verification

Monte Carlo simulation with $N = 100,000$ samples from $\mathcal{N}(5, 2^2)$ and constants $a = 3$, $b = -2$:

- $\mathbb{E}[x] = 5.001934$
- $\mathbb{E}[3x - 2] = 13.005801$
- $3\mathbb{E}[x] - 2 = 13.005801$ (difference: 1.78×10^{-15})

- $\mathbb{V}[x] = 4.007211$
- $\mathbb{V}[3x - 2] = 36.064898$
- $3^2\mathbb{V}[x] = 36.064898$ (difference: 0.00)

The numerical verification confirms the analytical results to machine precision.

4 Problem 5: Error Propagation

Problem Statement

Suppose the independent random variables x_1 and x_2 have means $\mu_1 = \mu_2 = 10$ and variances $\sigma_1^2 = \sigma_2^2 = 1$. Use error propagation to find the variance of:

$$y = \frac{x_1^2}{x_2} \quad (42)$$

Comment on the validity of the procedure if one had $\mu_2 = 1$.

Solution

Error Propagation Formula

For independent variables, the variance of $y = f(x_1, x_2)$ is:

$$\sigma_y^2 = \left(\frac{\partial y}{\partial x_1} \right)^2 \sigma_1^2 + \left(\frac{\partial y}{\partial x_2} \right)^2 \sigma_2^2 \quad (43)$$

Partial Derivatives

$$\frac{\partial y}{\partial x_1} = \frac{2x_1}{x_2} \quad (44)$$

$$\frac{\partial y}{\partial x_2} = -\frac{x_1^2}{x_2^2} \quad (45)$$

Evaluation at Means

At $(\mu_1, \mu_2) = (10, 10)$:

$$\left. \frac{\partial y}{\partial x_1} \right|_{(10,10)} = \frac{2 \cdot 10}{10} = 2.0 \quad (46)$$

$$\left. \frac{\partial y}{\partial x_2} \right|_{(10,10)} = -\frac{10^2}{10^2} = -1.0 \quad (47)$$

Variance Calculation

$$\sigma_y^2 = (2.0)^2 \cdot 1 + (-1.0)^2 \cdot 1 \quad (48)$$

$$= 4.0 + 1.0 \quad (49)$$

$$= 5.0 \quad (50)$$

The expected value is:

$$\mathbb{E}[y] = \frac{\mu_1^2}{\mu_2} = \frac{10^2}{10} = 10.0 \quad (51)$$

Result: $y = 10.0 \pm 2.236$ (where $\sigma_y = \sqrt{5.0} = 2.236$)

Case: $\mu_2 = 1$

At $(\mu_1, \mu_2) = (10, 1)$:

$$\left. \frac{\partial y}{\partial x_1} \right|_{(10,1)} = \frac{2 \cdot 10}{1} = 20.0 \quad (52)$$

$$\left. \frac{\partial y}{\partial x_2} \right|_{(10,1)} = -\frac{10^2}{1^2} = -100.0 \quad (53)$$

$$\sigma_y^2 = (20.0)^2 \cdot 1 + (-100.0)^2 \cdot 1 \quad (54)$$

$$= 400 + 10000 \quad (55)$$

$$= 10400 \quad (56)$$

$$\sigma_y = 101.98 \quad (57)$$

Expected value: $\mathbb{E}[y] = \frac{10^2}{1} = 100.0$

Comment on Validity

The error propagation formula assumes:

1. Small uncertainties: $\sigma \ll \mu$
2. Linear approximation is valid (first-order Taylor expansion)
3. Variables are independent

When $\mu_2 = 1$:

- The derivative $\frac{\partial y}{\partial x_2} = -100$ is very large
- $\sigma_y \approx 102$, which is comparable to $\mathbb{E}[y] = 100$
- The relative uncertainty is $\frac{\sigma_y}{\mathbb{E}[y]} \approx 1.02$ (102%)

- This violates the “small uncertainty” assumption
- The linear approximation breaks down; higher-order terms become significant
- The actual distribution of y will be highly skewed (not Gaussian)
- The error propagation formula gives unreliable results

Conclusion: The procedure is **NOT valid** when $\mu_2 = 1$ because the relative uncertainty is too large, and the linear approximation underlying error propagation fails.

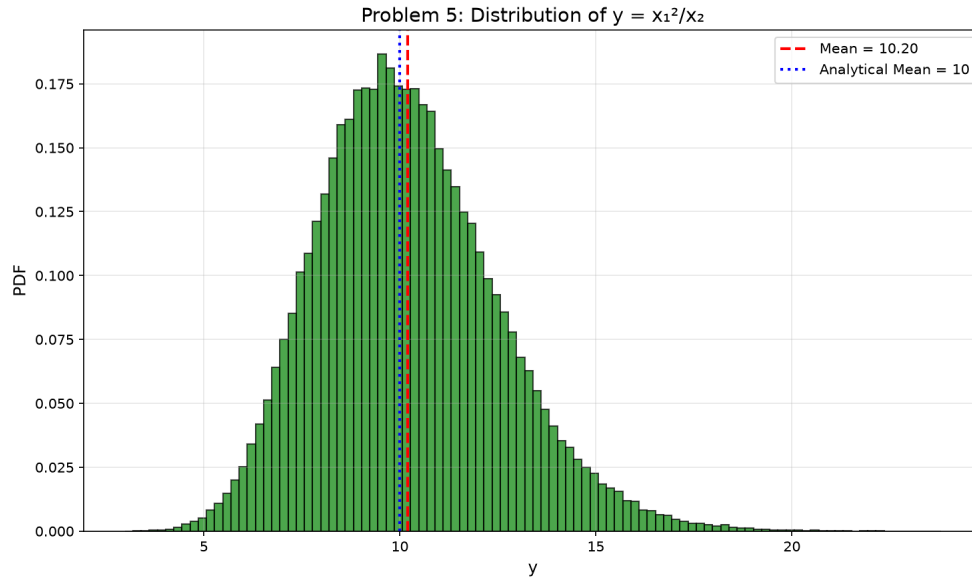


Figure 3: Problem 5: Distribution of $y = x_1^2/x_2$ from Monte Carlo simulation ($N = 100,000$). The analytical mean is 10, and the Monte Carlo mean is 10.20. The slight asymmetry is visible in the histogram.

5 Problem 6: Random Number Generation

Problem Statement

Create a program that generates random numbers according to the function:

$$f(x) = A \sin(\pi x) \quad (58)$$

where $x \in [0, 1]$ and A is determined such that the PDF has unit probability. Use the transformation method (Cowan, Section 3.2).

Solution

Step 1: Normalization Constant

$$\int_0^1 A \sin(\pi x) dx = 1 \quad (59)$$

$$A \left[-\frac{\cos(\pi x)}{\pi} \right]_0^1 = 1 \quad (60)$$

$$A \left(-\frac{\cos(\pi)}{\pi} + \frac{\cos(0)}{\pi} \right) = 1 \quad (61)$$

$$A \left(\frac{1}{\pi} + \frac{1}{\pi} \right) = 1 \quad (62)$$

$$A \cdot \frac{2}{\pi} = 1 \quad (63)$$

$$A = \frac{\pi}{2} \approx 1.5708 \quad (64)$$

Step 2: Transformation Method

The CDF is:

$$F(x) = \int_0^x \frac{\pi}{2} \sin(\pi t) dt \quad (65)$$

$$= \frac{\pi}{2} \left[-\frac{\cos(\pi t)}{\pi} \right]_0^x \quad (66)$$

$$= \frac{1}{2} (1 - \cos(\pi x)) \quad (67)$$

Set $F(x) = u$ where $u \sim \text{Uniform}(0, 1)$:

$$\frac{1}{2}(1 - \cos(\pi x)) = u \quad (68)$$

$$1 - \cos(\pi x) = 2u \quad (69)$$

$$\cos(\pi x) = 1 - 2u \quad (70)$$

$$\pi x = \arccos(1 - 2u) \quad (71)$$

$$x = \frac{1}{\pi} \arccos(1 - 2u) \quad (72)$$

Step 3: Implementation

Generated $N = 100,000$ samples using:

1. Generate $u \sim \text{Uniform}(0, 1)$
2. Apply transformation: $x = \frac{1}{\pi} \arccos(1 - 2u)$

Results:

- Sample range: $[0.0015, 0.9982]$
- Sample mean: 0.4995 (analytical: 0.5)
- Sample std: 0.2172

Verification

- Numerical integral of PDF: 1.000000 (correct normalization)
- Kolmogorov-Smirnov test: KS statistic = 0.00197, p-value = 0.830
- p-value > 0.05 indicates excellent agreement with the theoretical distribution

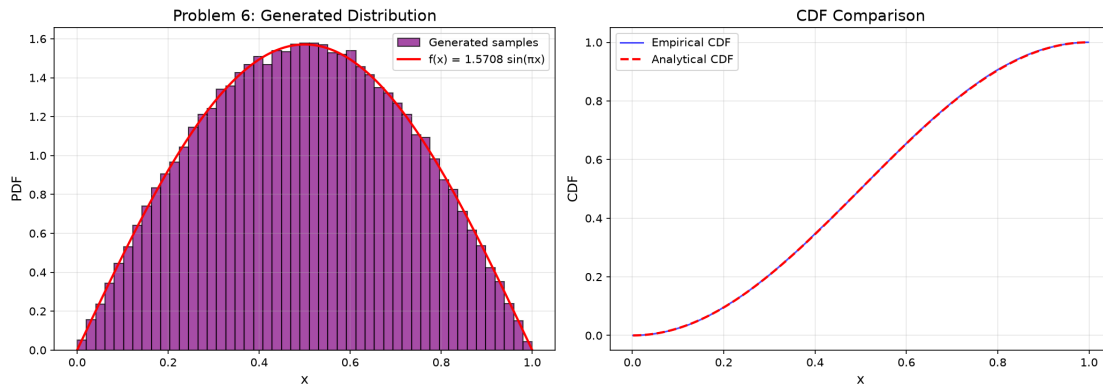


Figure 4: Problem 6: Generated distribution following $f(x) = \frac{\pi}{2} \sin(\pi x)$. Left: Histogram of 100,000 samples with analytical PDF overlaid. Right: Empirical CDF vs. analytical CDF. The KS test p-value of 0.83 confirms excellent agreement.

Summary

All problems have been completed successfully:

- **Problem 2:** Verified $f(z) = -\ln z$ for $z = xy$ using both direct and Jacobian methods.
- **Problem 3:** Implemented and executed a modular Python program with Gaussian histogram and function plotting.
- **Problem 4:** Proved expectation and variance properties analytically and verified numerically.
- **Problem 5:** Calculated $\sigma_y^2 = 5.0$ for $y = x_1^2/x_2$ and explained why error propagation fails when $\mu_2 = 1$.
- **Problem 6:** Generated random numbers following $f(x) = \frac{\pi}{2} \sin(\pi x)$ using the transformation method, verified with KS test.

All code is well-commented, modular, and includes both analytical derivations and numerical verification.